

**Memo 5: Non-Isothermal Fixed-Bed Reactor with Heat Transfer
Covering Chapter 6****Due Date: 06/05/2026****OVERVIEW**

Add energy balance to your Memo 4 reactor model to simulate realistic heat transfer with cooling. Analyze reactor hotspot temperature and predict dynamic stability using the “reactor gain” metric. Explore how inlet temperature and coolant temperature affect reaction selectivity and thermal stability.

PART 1: PYTHON MULTIPLE REACTION MODEL (50 pts)

Modify your Memo 4 Python model to include energy balance (dT/dW instead of isothermal):

Required deliverables:**1. Energy Balance Implementation (5 pts)**

- Derive dT/dW from Fogler Eq. 8-35 using your catalyst bulk density
- Document all assumptions: heat capacity values, overall heat transfer coefficient (U), coolant temperature
- Show energy balance equation clearly in memo with all variables defined
- References for U values and heat capacity data

2. Temperature Profile Analysis (20 pts)

- Plot temperature vs. catalyst weight at baseline conditions (T_{inlet} , $T_{coolant}$)
- Clearly mark the **hotspot** (maximum temperature location and value)
- Compare with isothermal operation: plot both T profiles on same graph
- Discussion: Where is hotspot? Why does it occur there?

3. Selectivity vs. Operating Conditions (20 pts)

- For constant conversion, evaluate selectivity at 3-4 different inlet temperatures
- For constant conversion, evaluate selectivity at 3-4 different coolant temperatures
- Create table: T_{inlet} vs. $T_{coolant}$ vs. $T_{hotspot}$ vs. Selectivity
- Discussion: How do operating conditions affect product selectivity?

4. Reactor Gain & Stability Analysis (5 pts)

- Calculate **reactor gain**: How much does $T_{hotspot}$ increase when $T_{coolant}$ increases by 1°C ?
- Formula: $\text{Gain} = \Delta T_{hotspot} / \Delta T_{coolant}$
- Stability criterion: Gain should be < 2 for dynamic stability
- Create table showing gain at different operating conditions
- **Discussion**: Is your design stable? If not, propose operating condition changes

PART 2: INTERACTIVE SLIDER VISUALIZATION (25 pts)

Create a React-based interactive tool (included in Python deliverable):

Build an interactive slider that allows real-time adjustment of:

- Inlet temperature (T_{inlet}): range 300–400 K
- Coolant temperature (T_{coolant}): range 250–350 K
- Overall heat transfer coefficient: range 0 - ???

For each slider position, display:

- Temperature profile plot (showing hotspot)
- Hotspot temperature value
- Conversion
- Selectivity to primary product
- Reactor gain
- Stability indicator (Green: $\text{Gain} < 2$ ✓ | Red: $\text{Gain} \geq 2$ ✗)

PART 3: POLYMATH VERIFICATION (Optional)

Set up equivalent energy balance model in POLYMATH with same kinetics, geometry, and cooling. Compare:

- Temperature profiles vs. Python
- Hotspot location and temperature
- Reactor gain calculations
- Selectivity results

Results should match within $\pm 1^\circ\text{C}$ hotspot temperature and ± 0.05 gain.

PART 4: ASPEN PLUS SIMULATION (Optional)

Use RPLUG reactor with energy balance and constant coolant temperature. Increase report intervals to see detailed temperature profile. Report:

- Temperature profile along reactor length
- Hotspot location and magnitude
- Reactor gain at baseline conditions
- Comparison with Python/POLYMATH results
- Effect of coolant temperature on selectivity

PART 5: REPORT QUALITY & COMMUNICATION (25 pts)

Professional presentation:

- All plots clearly labeled with axes, units, legend
- Writing is clear and concise
- All references properly cited

SUBMISSION CHECKLIST

- ☐ Energy balance derivation (dT/dW) with clear assumptions
- ☐ Temperature profile plot (non-isothermal vs. isothermal)
- ☐ Operating conditions table (T_{inlet} , T_{coolant} , T_{hotspot} , selectivity)
- ☐ Hotspot location identification and explanation
- ☐ Reactor gain calculation and stability assessment
- ☐ Discussion of stability: is design stable? How to improve?
- ☐ References for U , C_p values, assumptions
- ☐ Optional: Interactive slider visualization
- ☐ Optional: POLYMATH results & comparison
- ☐ Optional: Aspen Plus results & comparison

Format: Single PDF file, `Memo5\_GroupName.pdf`

Late submission: Less than 24 hrs: 50% credit only
More than 24hrs: 0% credit